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Computational Finance Project
Mathematical Engineering - Quantitative Finance

Asset Allocation

Group 3 - Exam A

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Contents

Contents i

Exercise 1 - Constrained and Robust Efficient Frontier ii

Exercise 2 - Black–Litterman Model iv

Exercise 3 - Diversification-Based Optimization viii

Exercise 4 - PCA and Conditional Value-at-Risk x

Exercise 5 - Personal Strategy xiii

Final Discussion xv

Exercise 1 - Constrained and Robust Efficient Frontier

1a. Classical Mean–Variance Efficient Frontier

In the first part of the analysis, the goal was to construct the mean–variance efficient frontier using in-sample data. The daily prices of the sixteen assets were first restricted to the period January 2018 to December 2022, from which daily log-returns were computed and then annualized to obtain the vector of expected returns and the covariance matrix. These empirical moments form the statistical input of the optimization problem. Before estimating the frontier, a large set of random long-only portfolios was generated to visualize the attainable risk–return region implied by the data. The efficient frontier was then built by solving a classical mean–variance problem under the full set of restrictions required by the exercise. In addition to full investment and long-only constraints we have:

$$w_i \leq 0.25 \qquad \sum_{i \in Def} w_i \leq 0.40 \qquad 0.45 \leq \sum_{i \in Cyc \cup Def} w_i \leq 0.75$$

Once the constrained frontier was estimated, two portfolios were extracted: Portfolio A is the Minimum Variance Portfolio, corresponding to the point of lowest volatility within the feasible set, Portfolio B is the Maximum Sharpe Ratio Portfolio, which maximizes expected return per unit of risk under the same constraints. The results of the two portfolios are shown in Table 1.

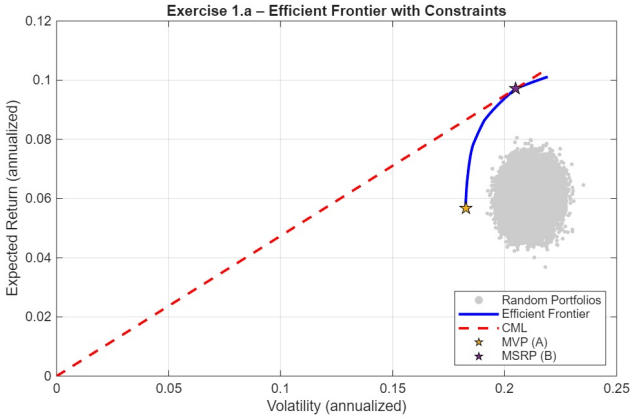


Figure 1.1: Efficient frontier under the imposed constraints alongside the Capital Market Line, with the two portfolios highlighted on the curve.

1b. Robust Frontier using a Resampling Approach

In this exercise, we implement a resampling approach to construct a robust efficient frontier, addressing the sensitivity of classical portfolio optimization to estimation errors in expected returns and covariances. To adhere precisely to the procedure described in the lecture slides, we slightly modify the original code: for each simulation, we generate a

synthetic dataset from a multivariate normal distribution using the historical daily mean and covariance, then recompute the sample mean and covariance from that simulated dataset. The efficient frontier is subsequently estimated using these sample moments, ensuring that each iteration reflects realistic estimation noise in both expected returns and covariances. For every simulation, we record portfolio risks, returns, Sharpe ratios, and weights. After completing all simulations, the average risks, returns, and weights across scenarios are used to construct the resampled frontier, which offers a more stable and reliable representation of the efficient set. From this robust frontier, we extract two representative portfolios: the Minimum Variance Portfolio, which minimizes overall volatility, and the Maximum Sharpe Ratio Portfolio, which maximizes risk-adjusted performance. Comparing the classical and resampled frontiers highlights how the resampling approach smooths extreme allocations and reduces the impact of estimation uncertainty, ultimately yielding more practical and dependable portfolio recommendations.

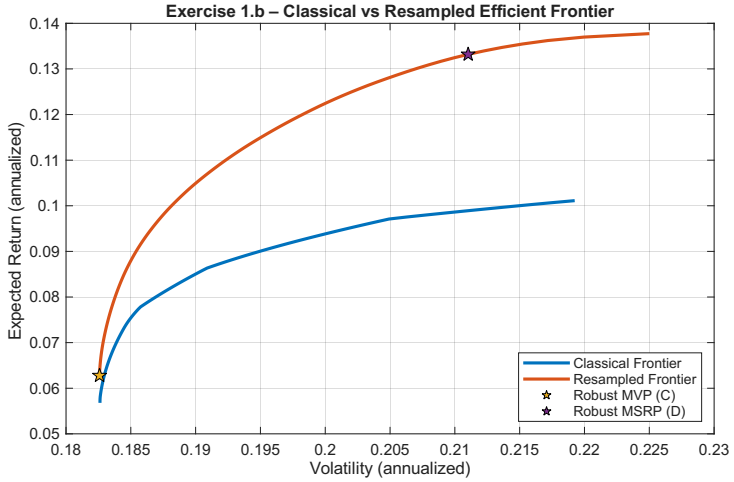


Figure 1.2: Comparison between the classical efficient frontier and the resampled one. The plot also highlights the robust Minimum Variance Portfolio (C) and the robust Maximum Sharpe Ratio Portfolio (D).

The weights of the Portfolios C and D are reported in the table below.

Asset	Portfolio A	Portfolio B	Portfolio C	Portfolio D
A1	0	0.2499	0	0.1529
A2	0.0958	0.0001	0.0906	0.0912
A3	0	0	0	0.0131
A4	0.2500	0.2500	0.2500	0.1294
A5	0.2367	0.2500	0.2231	0.1611
A6	0	0	0	0.0417
A7	0.1308	0	0.1289	0.0060
A8	0	0	0	0.0129
A9	0	0	0	0.0505
A10	0	0	0.0010	0.0376
A11	0	0	0.0040	0.0410
A12	0	0.2500	0.0032	0.1699
A13	0.2192	0	0.2138	0.0236
A14	0	0	0	0.0116
A15	0.0675	0	0.0863	0.0270
A16	0	0	0	0.0307

Table 1.1: Asset allocations across four portfolios (A–D). Each cell represents the proportion of the total portfolio invested in the corresponding asset.

Exercise 2 - Black–Litterman Model

2a. Equilibrium Returns

Starting from the `weights_csv` market-cap weighted portfolio, the equilibrium returns are computed using the standard Black–Litterman formula:

$$\mu_{\text{eq}} = \lambda \, \Sigma \, w_{\text{MKT}}$$

where Σ is the in-sample covariance matrix of asset returns, and λ is the risk aversion coefficient. We choose $\lambda = 1.2$, a commonly used value in empirical asset allocation exercises. The resulting vector μ_{eq} serve as a neutral starting point for portfolio optimization before incorporating any investor views and are shown in Table 1.2.

2b. Introduction of the Views

Three specific views are introduced to adjust the equilibrium returns: the group of cyclical assets is expected to outperform the defensive group by 2% annualized, Asset 3 is expected to outperform Asset 11 by 1% annualized, and Asset 7 is expected to outperform the average of defensive assets by 0.5% annualized. These views are combined into matrices P and q to be used in the Black–Litterman posterior calculation. The uncertainty associated with each view is modeled in the diagonal matrix Ω.

2c. Posterior Expected Returns

Using the Black–Litterman approach, the posterior expected returns and the adjusted covariance matrix are computed as:

$$\mu_{BL} = \underbrace{\Big((\tau\Sigma)^{-1} + P^{\top}\Omega^{-1}P \Big)^{-1}}_{\Sigma_{BL}} \Big((\tau\Sigma)^{-1}\mu_{\text{eq}} + P^{\top}\Omega^{-1}Q \Big)$$

where $\tau = 1/N_{obs}$. The efficient frontier is then constructed under standard constraints, i.e., full investment and no short-selling. The Black–Litterman frontier is compared with the classical (mean-variance) frontier based on the original equilibrium returns.

Asset	Equilibrium Return (μ_{eq})	Market-cap Weight	BL Posterior Return (μ_{BL})
A1	0.0643	0.1000	0.0675
A2	0.0410	0.0583	0.0385
A3	0.0560	0.0583	0.0586
A4	0.0365	0.1000	0.0373
A5	0.0433	0.0583	0.0436
A6	0.0620	0.0300	0.0697
A7	0.0510	0.0300	0.0541
A8	0.0574	0.1000	0.0645
A9	0.0691	0.1000	0.0872
A10	0.0574	0.1000	0.0649
A11	0.0506	0.0300	0.0503
A12	0.0539	0.0300	0.0560
A13	0.0523	0.0300	0.0575
A14	0.0542	0.0583	0.0578
A15	0.0438	0.0583	0.0441
A16	0.0577	0.0583	0.0601

Table 1.2: Equilibrium returns (μ_{eq}), market-cap weighted portfolio, and Black–Litterman posterior returns (μ_{BL}) for the 16 assets.

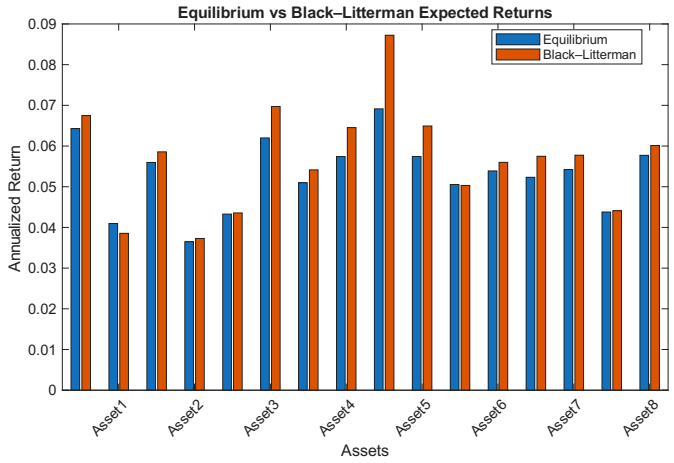


Figure 1.3: Comparison between Annualized Returns of each asset according to the equilibrium and the Black-Litterman model

2d. Portfolios E & F

From the posterior frontier, we now select again the **MVP (Portfolio E)** and the **MRSP (Portfolio F)** which reflect the adjustments induced by the investor's views and are shown in the table below.

Asset	Portfolio E (MVP BL)	Portfolio F (MSRP BL)
A1	0.0000	0.1644
A2	0.0000	0.0000
A3	0.0000	0.0000
A4	0.7346	0.1282
A5	0.2495	0.0000
A6	0.0000	0.0238
A7	0.0159	0.0753
A8	0.0000	0.1929
A9	0.0000	0.2162
A10	0.0000	0.1992
A11	0.0000	0.0000
A12	0.0000	0.0000
A13	0.0000	0.0000
A14	0.0000	0.0000
A15	0.0000	0.0000
A16	0.0000	0.0000

Table 1.3: Weights of Portfolio E and Portfolio F under the Black–Litterman approach.

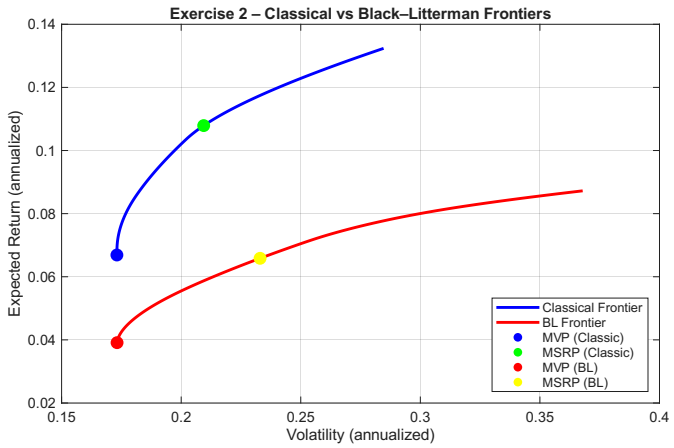


Figure 1.4: Comparison of the classical efficient frontier and the Black–Litterman frontier. The plot highlights the MVP and MSRP for both approaches.

2e. Effects of the Investor's Views

The introduction of investor views through the Black–Litterman framework produces a substantial shift in both the structure and diversification of the optimal portfolios. This becomes evident when comparing the classical solutions to their Black–Litterman counterparts. The Black–Litterman Minimum Variance Portfolio (Portfolio E) is highly concentrated but differs meaningfully from the classical allocation. Under the posterior inputs, the MVP places approximately 73 % of total weight in Asset 4, 25 % in Asset 5, and a small residual allocation in Asset 7. All remaining assets receive zero weight. This indicates that, given the adjusted covariance matrix, those three assets jointly form the lowest-risk combination once market equilibrium information and views are incorporated. Notably, the inclusion of views does not necessarily diversify the MVP; instead, it slightly reallocates toward assets that benefit from reduced perceived risk or favorable posterior correlations.

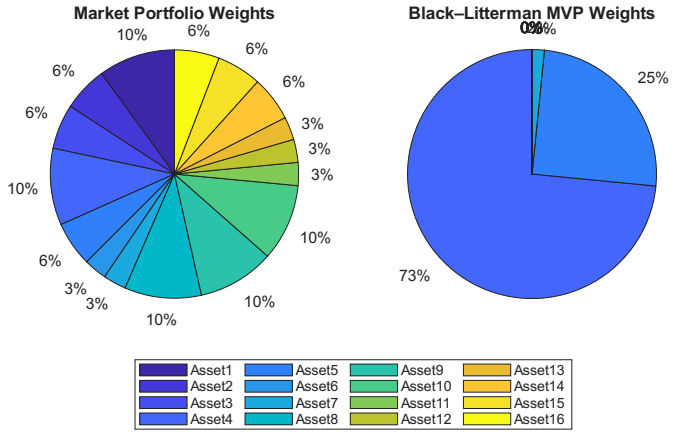


Figure 1.5: Comparison of asset allocations: left, the market-cap weighted portfolio; right, the Black–Litterman MVP. The legend shows the individual assets.

The Black–Litterman MSRP (Portfolio F) displays a dramatic increase in diversification relative to the classical solution. The classical MSRP (with **no constraints** on the

weights) invests almost exclusively in two assets (Asset 1 and 5), which reflects the instability and extreme concentration typical of mean–variance optimization based purely on sample returns. After introducing the views, the BL-MSRP spreads its weights across seven different assets (Assets 1, 4, 6, 7, 8, 9, and 10), none of which exceed roughly 22 % of the portfolio. This broader allocation results from two mechanisms: (i) posterior expected returns are pulled toward equilibrium, reducing estimation noise, and (ii) the imposed views explicitly increase the relative attractiveness of certain assets (e.g., Asset 7 and Cyclicals) while penalizing others (e.g., Defensive assets in relative terms). Assets directly targeted by the views—most notably Asset 7 (expected to outperform Defensives) and the Cyclical group—receive non-trivial weights, whereas Asset 11, which is expected to underperform Asset 3, remains excluded. The sizable allocations to Assets 8, 9, and 10 arise from the stabilizing effect of the BL posterior, which balances prior information and views to produce smoother, less extreme expected return estimates. Overall, the Black–Litterman approach yields more diversified and economically interpretable portfolios, particularly for the Sharpe-maximizing solution. While the MVP remains concentrated due to the inherent structure of minimum-variance optimization, the MSRP shows clear improvements in both stability and portfolio breadth.

Exercise 3 - Diversification-Based Optimization

3a. Construction of the Portfolios

Using the in-sample covariance matrix, we construct two portfolios: the Maximum Diversification Ratio **(G)** and the Equal Risk Contribution **(H)**. Both optimizations impose full investment, no short-selling, a 20% cap per asset, and a minimum 15% exposure to each macro group.

Portfolio G maximizes the diversification ratio:

$$DR(w) = \frac{\sum_{i=1}^{N} w_i \sigma_i}{\sqrt{w^\top \Sigma_{\text{daily}}\, w}},$$

where w is the portfolio weight vector, σ_i are individual volatilities, and Σ_{daily} is the covariance matrix of daily returns. The resulting allocation is highly selective: weights concentrate on Assets 2 and 9 (both at 20%) and on Assets 3, 4 and 7, while many assets receive zero weight, indicating limited contribution to diversification under the empirical correlation structure.

Portfolio H enforces equal risk contributions by minimizing:

$$\sum_{i=1}^{N} \left(\frac{w_i (\Sigma_{\text{daily}} w)_i}{w^\top \Sigma_{\text{daily}} w} - \frac{1}{N} \right)^2,$$

so that each asset contributes the same fraction of total portfolio volatility. Under the same constraints, ERC delivers a much more uniform allocation, with weights typically between 5% and 8%.

Asset	Portfolio G (Max DR)	Portfolio H (ERC)
1	0.0000	0.0506
2	0.2000	0.0759
3	0.1353	0.0580
4	0.1956	0.0870
5	0.0444	0.0738
6	0.0000	0.0530
7	0.1895	0.0627
8	0.0000	0.0576
9	0.2000	0.0505
10	0.0000	0.0578
11	0.0351	0.0623
12	0.0000	0.0599
13	0.0000	0.0625
14	0.0000	0.0600
15	0.0000	0.0725
16	0.0000	0.0560

Table 1.4: Portfolio weights for the Maximum Diversification Ratio (G) and Equal Risk Contribution (H).

3b. Comparison of the Portfolios

Metric	G (MDR)	H (ERC)	EW
Diversification Ratio	1.2514	1.1541	1.1533
Volatility (daily)	0.01272	0.01291	0.01320
Sharpe Ratio	0.01309	0.01841	0.01808
Effective N. Assets	5.69	15.62	16.00

Table 1.5: Diversification ratio, volatility, Sharpe ratio and effective number of assets for portfolios G, H and EW.

Portfolio G achieves the highest diversification ratio and the lowest volatility, confirming that its concentrated structure exploits the underlying correlation matrix efficiently. However, its effective number of assets (ENA = 5.69) is low, reflecting limited diversification by weights. Portfolio H, instead, achieves a much broader allocation (ENA circa 15.6) while maintaining low volatility and delivering the highest Sharpe ratio, suggesting superior risk-adjusted performance. The equally weighted benchmark is fully diversified by construction but exhibits the lowest diversification ratio and the highest volatility among the three.

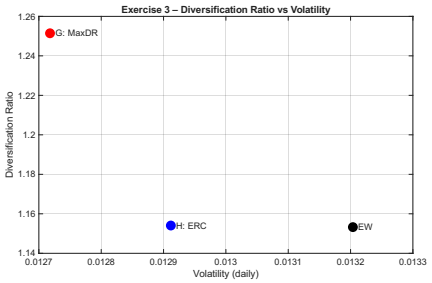


Figure 1.6: Daily volatility vs. diversification ratio for portfolios G, H, and EW.

The next figure reports the allocations across Cyclical, Neutral, and Defensive groups. Portfolio G naturally tilts toward the groups that contribute more to reducing aggregate risk given the covariance structure, whereas Portfolio H maintains a more balanced exposure profile, remaining close to the equally weighted allocation due to its risk–parity formulation.

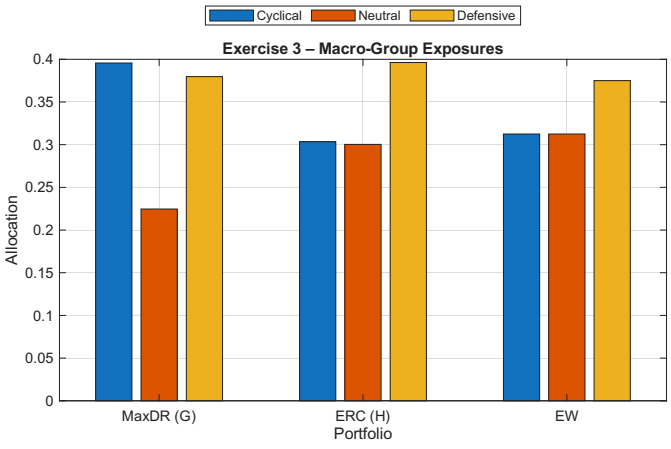


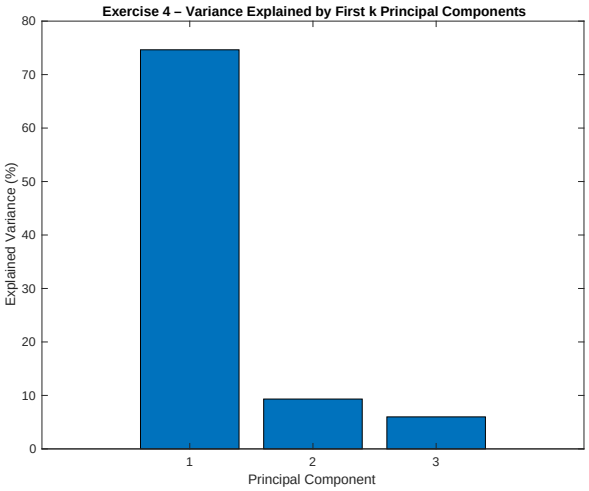
Figure 1.7: Macro-group exposures for Portfolios G, H, and the Equally Weighted benchmark.

Exercise 4 - PCA and Conditional Value-at-Risk

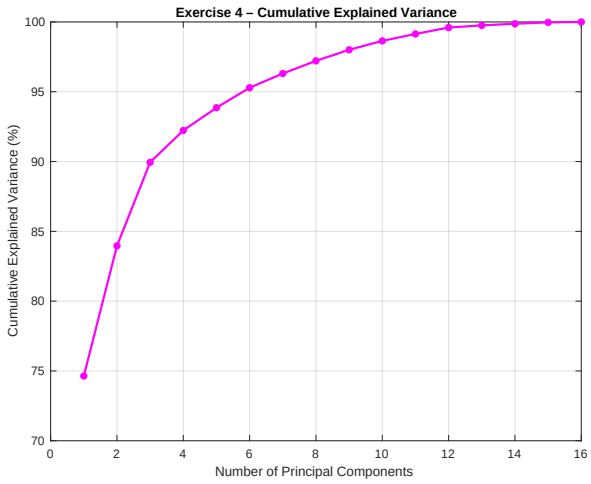
4a. Principal Component Analysis

The goal of this exercise is to exploit Principal Component Analysis (PCA) to obtain a filtered estimate of the covariance matrix and to design portfolios that explicitly control their exposure to the main sources of risk.

PCA is applied to the covariance matrix Σ, yielding eigenvalues λ_i and eigenvectors v_i. The eigenvalues measure the variance explained by each principal component, while the corresponding eigenvectors represent orthogonal risk factors.



(a) Figure 1.8 reports the fraction of total variance explained by the first k-components.



(b) Figure 1.8 shows the Cumulative Explained Variance.

The first component alone captures a large portion of the total variance (more than 70%) and can be interpreted as a broad market factor. The first k components are retained so that the cumulative explained variance exceeds 85%; the remaining components are treated as noise. Then, using only the retained components, we reconstruct a filtered covariance matrix

$$\Sigma_{\text{PCA}} = V_k \Lambda_k V_k',$$

where V_k stacks the first k eigenvectors and Λ_k is the diagonal matrix of the associated eigenvalues. This matrix preserves the main dependence structure while reducing estimation error.

4b. Portfolio I – PCA Max–Sharpe Portfolio

Portfolio I is defined as the maximum Sharpe ratio portfolio built using the filtered covariance matrix Σ_{PCA}. The optimisation problem maximises

$$\frac{\mu' w}{\sqrt{w' \Sigma_{\text{PCA}} w}}$$

subject to full investment, long–only constraints and a 25% cap per asset. To reduce sensitivity to the dominant market factor, we also constrain the exposure to the first principal component. Denoting by v_1 the eigenvector associated with the largest eigenvalue, the portfolio weights satisfy

$$w' v_1 \leq c,$$

with c chosen to keep the loading on PC1 moderate. In this way, Portfolio I concentrates risk on secondary components rather than on the global market mode, promoting diversification across more specific sources of variation. The optimal allocation tends to be reasonably spread across assets, with limited concentration thanks to the box constraints. Assets that load positively on attractive principal components receive larger weights, while those associated with less rewarded factors are down-weighted.

4c. Portfolio J – CVaR–minimising Portfolio

Portfolio J is constructed by explicitly targeting tail risk, measured by the 5% Conditional Value-at-Risk (CVaR). Let r_t denote the vector of asset returns and $L_t(w) = -w' r_t$ the portfolio loss. For a confidence level $\alpha = 0.95$, the CVaR is

$$\text{CVaR}_{0.95}(w) = \mathbb{E}[\, L_t(w) \,|\, L_t(w) \geq \text{VaR}_{0.95}(w)] \,.$$

Using the Rockafellar–Uryasev formulation, we obtain a linear programming problem that delivers the weights minimising CVaR under full-investment and long-only constraints. Initially, a volatility cap of 15% (annualised, computed with Σ_{PCA}) is imposed as a nonlinear constraint. However, testing the minimum-variance portfolio under the same bounds shows that the lowest achievable volatility in this universe is above 15% (circa 17%). The volatility cap is therefore infeasible and the final CVaR-minimising solution is obtained without it, still respecting the long-only and 25% per-asset limits. The resulting Portfolio J naturally tilts toward assets that contribute less to large negative returns, often favouring more defensive exposures. Its expected return is typically lower than that of Portfolio I, but the distribution of losses in the left tail is noticeably improved.

Figure 1.9 reveals interesting differences in the behaviour of the two curves. In particular: Portfolio I (PCA Max Sharpe): higher growth overall but larger drawdowns, especially around 2020. Portfolio J (CVaR + Vol Cap): smoother, less volatile path with smaller losses in crises, but lower final wealth.

Asset	Portfolio I	Portfolio J
1	0.2499	0.0000
2	0.0004	0.1242
3	0.0000	0.0000
4	0.2495	0.2500
5	0.2500	0.2500
6	0.0000	0.0000
7	0.0000	0.1258
8	0.0000	0.0000
9	0.0000	0.0000
10	0.0000	0.0000
11	0.0000	0.0000
12	0.2498	0.0000
13	0.0000	0.0000
14	0.0001	0.0000
15	0.0001	0.2500
16	0.0001	0.0000

Table 1.6: Portfolio weights for Portfolio I and Portfolio J.

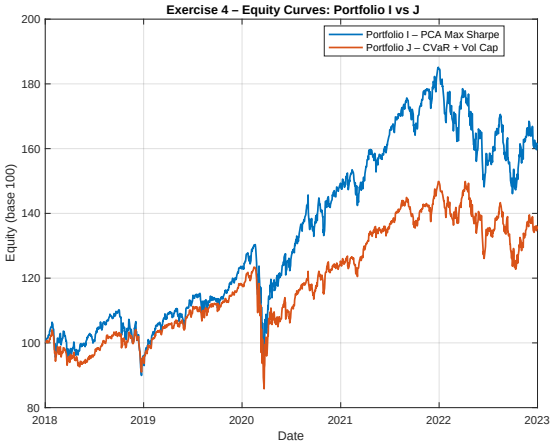


Figure 1.9: Comparison between the equity curves (value over time, starting at 100) from 2018 to 2023 for two strategies

4d. Comparison of Portfolios I and J

To assess the two strategies, we compute key performance indicators: annualised return and volatility, Sharpe ratio, maximum drawdown and 5% CVaR. Portfolio I, driven by mean–variance considerations, achieves the highest expected return and Sharpe ratio, but experiences deeper drawdowns and more severe tail losses. Portfolio J, by construction, reduces CVaR and maximum drawdown, at the cost of a more modest performance in benign market phases.

Exercise 4			
	CVaR_5pct	**AnnVol**	**MaxDD**
PCA MaxSharpe (I)	-0.031875	0.20442	-0.28859
PCA CVaR (J)	-0.027668	0.17933	-0.30435

Table 1.7: Comparison of tail risk, volatility and maximum drawdown for Portfolios I and J.

Overall, Exercise 4 illustrates the trade-off between factor-based optimisation (via PCA) and tail-risk control (via CVaR). PCA helps to obtain a cleaner covariance estimate and to limit exposure to the dominant market factor, while CVaR optimisation allows the investor to explicitly shape the behaviour of the portfolio in extreme scenarios.

Exercise 5 - Personal Strategy

5a. Trend–Following View-Based Strategy

In this chapter we describe the construction of the directional views used inside the Black–Litterman model. Instead of relying on discretionary macro opinions or sector rotations, we adopt a fully mechanical and replicable approach. The views originate from a simple comparison between market prices and their moving averages.

More precisely, for each asset we compute two technical signals:

- the **50-day moving average (MA50)**, representing short-to-medium term trend;
- the **200-day moving average (MA200)**, representing long-term trend.

The rule is intentionally straightforward:

An asset generates a positive view if its price is above both moving averages; a negative view if it is below both. Otherwise, we take a neutral stance.

This mechanism embeds a trend-following logic: the larger and more persistent the price deviation from its moving averages, the stronger the perceived drift. The motivation is that momentum tends to persist, and moving averages provide a transparent, non-discretionary way to identify it.

View Construction and Integration into Black–Litterman

For each asset i, we define the view q_i as:

$$q_i = \begin{cases} +v & \text{if } P_i > MA50_i \text{ and } P_i > MA200_i, \\ -v & \text{if } P_i < MA50_i \text{ and } P_i < MA200_i, \\ 0 & \text{otherwise,} \end{cases}$$

where v is the view intensity (set to 3% in our application).

To combine the views with the equilibrium (market-cap–weighted) portfolio, each view is formulated as a relative opinion: we construct for each asset a P row that represents the deviation between the asset and the market portfolio. This ensures the model incorporates both the expected direction and its comparison against the global benchmark.

Results

After incorporating the trend-following views into the Black–Litterman posterior distribution, we select the Maximum Sharpe portfolio and evaluate it against the market-cap portfolio for comparison

The table below reports the main performance statistics:

Statistic	BL Strategy	Market-Cap
Annualized Return	19.94%	17.84%
Annualized Volatility	11.31%	11.12%
Sharpe Ratio	1.76	1.57
Max Drawdown	10.69%	10.12%
Calmar Ratio	1.86	1.72

Overall, the trend-following Black–Litterman strategy delivers a more favorable risk–reward profile, mainly due to a higher Sharpe and a higher Calmar ratio. This indicates a more stable capital trajectory, particularly during periods of market stress.

Conclusions

We compare the portfolio's performance against the market-cap benchmark.

The following plot displays the cumulative performance of both strategies. the trend-following BL strategy shows a clear improvement over the market-cap allocation, confirming that the signals extracted from moving averages retain predictive power and translate into more resilient performance dynamics.

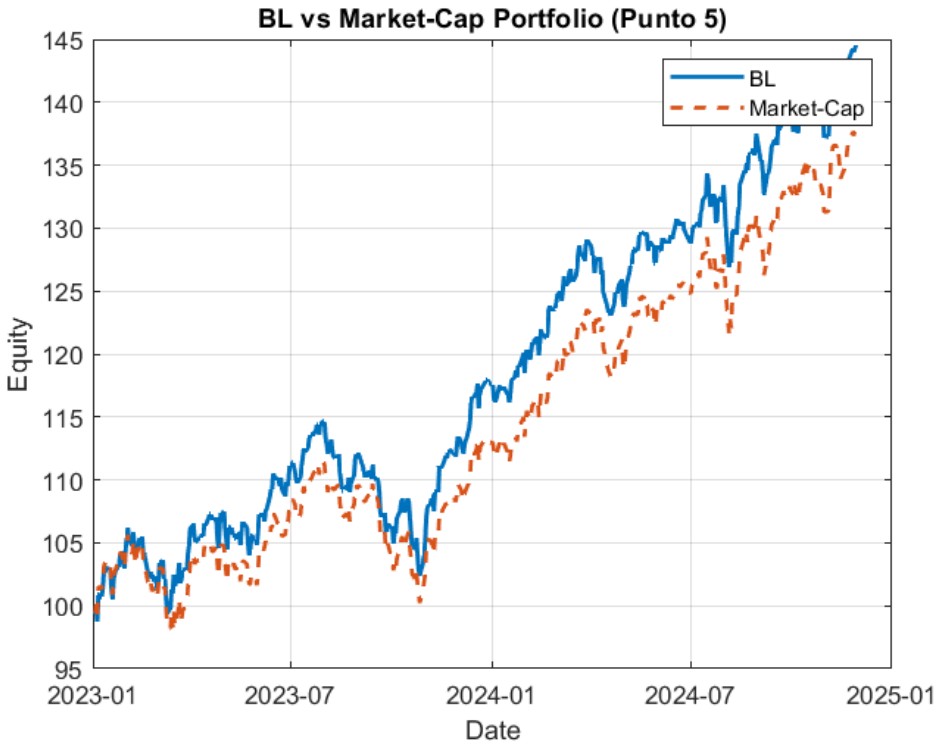


Figure 1.10: Out-of-sample cumulative performance comparison between our strategy and the market-cap benchmark.

Final Discussion

Out-of-sample performance of portfolios A–J

In this section we compare the out-of-sample (OOS) behaviour, from January 2023 to December 2024, of all portfolios constructed in the previous exercises. For each portfolio we compute annualised return, annualised volatility, the Sharpe ratio, the maximum drawdown and the Calmar ratio.

Portfolio	AnnReturn	AnnVol	Sharpe	MaxDrawdown	Calmar
A – MVP	0.131 71	0.098 361	1.3391	0.104 69	1.2582
B – Max Sharpe	0.195 95	0.113 360	1.7286	0.085 766	2.2847
C – Robust MVP	0.131 92	0.098 191	1.3435	0.104 49	1.2625
D – Robust Max Sharpe	0.176 04	0.108 040	1.6294	0.093 927	1.8742
E – BL MVP	0.072 10	0.098 072	0.7352	0.116 72	0.6177
F – BL Max Sharpe	0.174 28	0.120 720	1.4437	0.097 478	1.7879
G – Max Diversification	0.157 44	0.110 310	1.4273	0.096 806	1.6264
H – Equal Risk Contribution	0.177 69	0.109 670	1.6203	0.101 79	1.7457
I – PCA Max Sharpe	0.195 98	0.113 370	1.7287	0.085 778	2.2848
J – PCA CVaR	0.108 19	0.096 711	1.1187	0.102 66	1.0538

Table 1.8: Out-of-sample performance (Jan 2023 – Dec 2024) of portfolios A–J.

The best strategies in terms of annualised return are the Sharpe–oriented portfolios: **B** (Max Sharpe) and **I** (PCA Max Sharpe) deliver almost 20% per year, followed by **D** (Robust Max Sharpe) and **F** (BL Max Sharpe). On the other side, the most conservative configuration, **E** (BL MVP), records the lowest return (about 7%).

In terms of volatility, the most defensive allocations are the minimum variance portfolios **A** and **C**, together with the Black–Litterman MVP **E**, all around 9.8% annualised risk. However, this reduction in volatility does not necessarily translate into superior risk–adjusted performance: the highest Sharpe ratios are achieved by the more aggressive specifications **B**, **I**, **D** and **H**, all above 1.6, with **B** and **I** reaching approximately 1.73.

Maximum drawdowns are also relatively contained for the best performers. Portfolios **B** and **I** combine high returns with the smallest drawdowns in the set (around 8.6%), leading to the largest Calmar ratios (about 2.28). Overall, the OOS evidence suggests that Sharpe–oriented and PCA–filtered portfolios dominate the more conservative minimum–variance configurations over this particular market phase.

To complement the numerical analysis, the following figure reports the cumulative performance of all portfolios A-J over the OOS period.

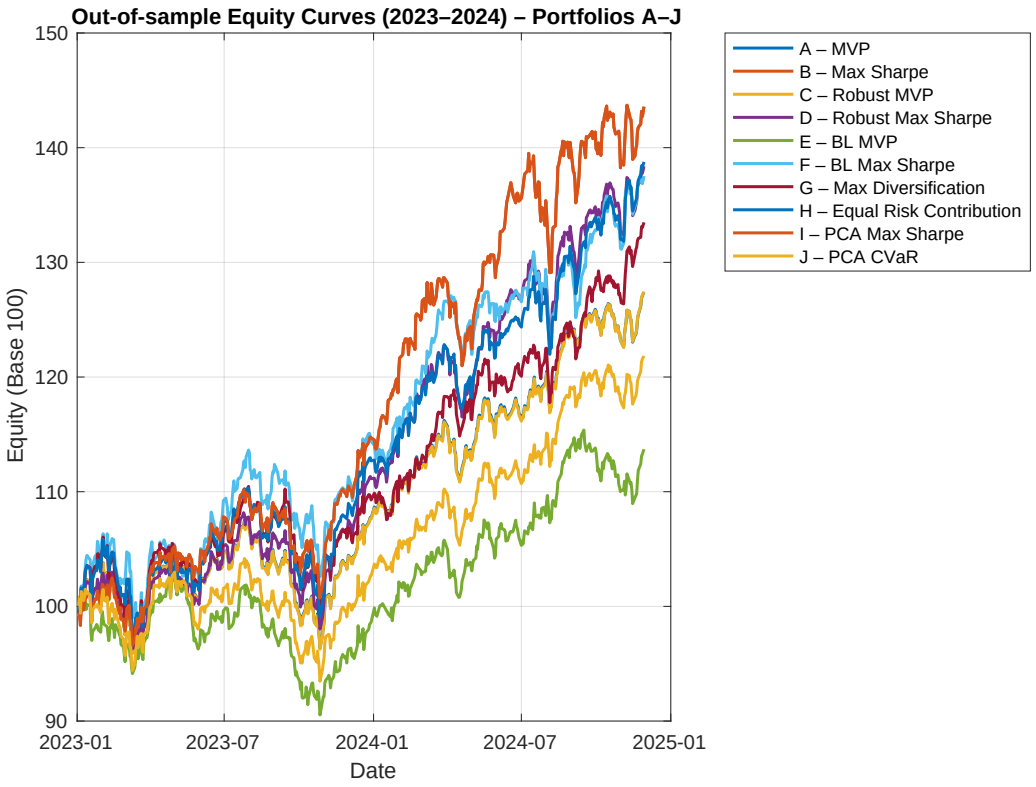


Figure 1.11: Out-of-sample cumulative performance (Jan 2023 – Dec 2024) of portfolios A–J.

Risk-adjusted performance and robustness

The comparison between classical MVP and MSRP solutions and their robust counterparts highlights how different modelling choices affect risk–adjusted performance.

Classical vs. resampled efficient frontier (A vs. C, B vs. D)

For the minimum variance portfolios, **A** and **C** are almost indistinguishable: they exhibit nearly identical returns, volatility and Sharpe ratios. This indicates that, for this data set, resampling the inputs does not improve the behaviour of the MVP.

For the maximum Sharpe allocations, the situation is slightly different. **B** (Max Sharpe) achieves both higher return and higher Sharpe ratio than its resampled version **D** (Robust Max Sharpe), at the cost of only a modest increase in volatility. Hence, in this case the resampling procedure actually dampens performance more than it enhances robustness. The classical solution remains dominant in the OOS analysis.

Classical vs. Black–Litterman (E, F vs. A, B)

The Black–Litterman minimum variance portfolio **E** turns out to be the worst performer in the entire set. Although its volatility is low, the combination of market equilibrium and views produces an allocation that is too defensive and underperforms substantially in terms of both return and Sharpe.

By contrast, the Black–Litterman maximum Sharpe portfolio **F** displays a more attractive trade–off: its return is only slightly below that of the most aggressive portfolios,

while volatility and drawdowns are moderate. This suggests that, when applied to Sharpe–seeking strategies, the Black–Litterman mechanism can successfully stabilise the solution and produce more robust risk–return profiles.

Diversification-based portfolios (G, H)

The diversification–oriented allocations **G** (Max Diversification) and **H** (Equal Risk Contribution) offer a very desirable compromise between risk and performance. Both exhibit volatility around 11%, significantly lower than the most aggressive Sharpe–driven portfolios, while maintaining Sharpe ratios above 1.4. In particular, **H** delivers one of the highest Sharpe ratios in the entire sample (1.62) with limited drawdowns, confirming the well–known empirical robustness of ERC solutions.

PCA-based portfolios (I, J)

The PCA–based maximum Sharpe portfolio **I** is among the most successful allocations: it matches the return and Sharpe ratio of the classical Max Sharpe portfolio **B**, but benefits from a covariance matrix where noise is filtered out through the principal component reconstruction. The resulting OOS behaviour is extremely attractive: high return, controlled volatility and very small drawdowns.

On the other hand, the CVaR–oriented portfolio **J** is more conservative: its volatility is among the lowest in the set but return and Sharpe are considerably reduced. The additional CVaR constraint is therefore effective in limiting tail risk, but at the cost of a sizable performance penalty in this sample.

Role of Cyclical, Neutral and Defensive exposures

The different out-of-sample behaviours can be largely explained by the macro–group exposures constructed in the earlier exercises. Portfolios with strong tilts towards **Cyclical** assets (notably **B**, **D**, **F** and **I**) achieve the highest OOS returns and Sharpe ratios, indicating that the 2023–2024 period was favourable to risk–on segments of the universe. Despite their pro–cyclical stance, these strategies do not exhibit extreme drawdowns, which suggests that the realised market environment rewarded cyclical risk rather than penalising it.

Conversely, more defensive allocations,such as the minimum–variance portfolios **A** and **C**, and especially the Black–Litterman MVP **E**,carry larger weights in **Neutral** and **Defensive** groups. While this naturally reduces volatility, it also limits upside participation and leads to noticeably lower risk–adjusted returns in the considered OOS window.

The diversification–based portfolios **G** and **H** stand in between these two extremes: they maintain balanced exposures across Cyclical, Neutral and Defensive assets, which translates into good Sharpe ratios, moderate volatility and controlled drawdowns. In particular, the ERC allocation **H** manages to combine well–balanced macro–group weights with one of the best risk–adjusted profiles in the entire set.

Discussion of the Out-of-Sample Performance of our BL-MA Strategy

The Black–Litterman Moving-Average strategy implemented in Exercise 5 delivers a weaker out-of-sample performance than the market-cap benchmark during 2023–2025. Its annualized return (4.09%), Sharpe ratio (0.263), and Calmar ratio (0.273) are all below the corresponding metrics of the passive portfolio, and the strategy exhibits a comparable maximum drawdown (14.9%). This section discusses the main reasons behind this outcome.

Portfolio	AnnReturn	AnnVol	Sharpe	MaxDrawdown	Calmar
BL_MA_Strategy	0.040924	0.15568	0.26287	0.14984	0.27311
MarketCap	0.048442	0.16006	0.30265	0.16042	0.30196

Table 1.9: 2025 OOS performances (updated to 08-Dec-2025)

The BL_MA strategy delivers an annualized return of approximately 4.1%, which is lower than the market portfolio return of 4.8%. This underperformance is consistent with the conservative nature of the strategy, which combines regime-based tilting with a constrained optimization framework. While the strategy does not outperform the benchmark in terms of raw returns, it exhibits a slightly lower annualized volatility (15.6% versus 16.0%), reflecting a more controlled risk exposure.

From a risk-adjusted perspective, the Sharpe ratio of the BL_MA strategy (0.26) remains below that of the market portfolio (0.30). This indicates that, over the 2025 window, the reduction in volatility achieved by the strategy was not sufficient to compensate for the loss in expected return. As a result, the market portfolio maintains superior efficiency in mean–variance terms.

However, the drawdown-based metrics provide a more nuanced picture. The BL_MA strategy achieves a lower maximum drawdown (approximately 15.0%) compared to the market portfolio (16.0%). This improvement is consistent with the defensive tilting mechanism embedded in the strategy, which shifts exposure toward defensive assets during bearish regimes identified through moving-average signals. Consequently, the strategy offers improved downside protection relative to the benchmark.

The Calmar ratio further confirms this behavior. Although the market portfolio still dominates with a higher Calmar ratio, the BL_MA strategy shows a non-negligible improvement compared to its own 2023–2024 out-of-sample performance, suggesting increased stability when extending the dataset and updating the Black–Litterman inputs using a longer historical window.

Overall, the 2025 out-of-sample results highlight a key trade-off inherent in the proposed strategy. The BL_MA approach focuses on controlling drawdowns and adjusting to market regimes, even if that means giving up some gains when markets are strong. While this leads to underperformance relative to a fully invested market portfolio in terms of returns and Sharpe ratio, it provides smoother equity dynamics and reduced tail risk. These characteristics make the strategy more suitable for risk-averse investors or as a defensive overlay rather than as a pure return-maximizing allocation.

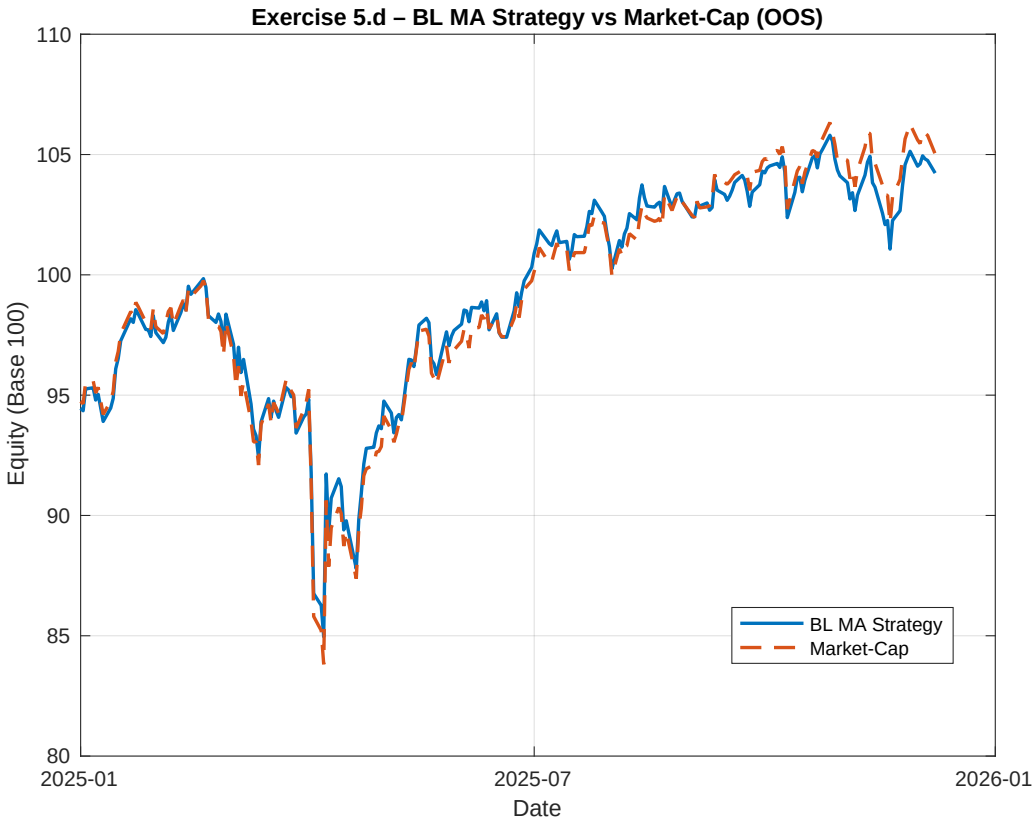


Figure 1.12: Out-of-sample cumulative performance (Jan 2025 – Dec 2025) of our strategy